

HTML

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(HyperText Markup Language)

1. Introduction

HTML stands for HyperText Markup Language. It is the standard language used to create web pages. It describes the structure of a web page using markup.

2. Basic Structure

★ Explanation :

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello World!</h1>
    <p>This is my first
      web page.</p>
  </body>
</html>
```

- **<!DOCTYPE html>** : Declares the document type.
- **<html> </html>** : Root element of HTML page.
- **<head> </head>** : Contains meta information about the page.
- **<title> </title>** : Sets the title of the page (shown on browser tab).
- **<body> </body>** : Contains the content of the page.
- **<h1> </h1>** : Heading (level 1).
- **<p> </p>** : Paragraph.

3. Common HTML Tags

<u>Tag</u>	<u>Description</u>	<u>Example</u>
<h1> to <h6>	Defines headings	<h1>Heading 1</h1>
<p>	Defines a paragraph	<p>This is a paragraph.</p>

	Line break	Hello
 World
<hr>	Horizontal rule	<hr>
<a>	Defines a link	 Click here
	Embeds an image	
	Bold text	Bold Text
<i>	Italic text	<i>Italic Text</i>
<u>	Underlined text	<u>Underlined Text</u>

4. Comments in HTML

Comments are used to give notes or explanations in code.

```
<!-- This is a comment -->
```

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5. Common Formatting Tags

Tag	Description	Example
<code></code>	Bold text	<code>This is bold</code>
<code><i></code>	Italic text	<code><i>This is italic</i></code>
<code><u></code>	Underlined text	<code><u>This is underlined</u></code>
<code></code>	Important text (bold)	<code>Important</code>
<code></code>	Emphasized text (italic)	<code>Emphasized</code>
<code><small></code>	Smaller text	<code><small>Small text</small></code>

6. Lists in HTML

HTML supports two types of lists: Ordered List (`<ol>`) and Unordered List (`<ul>`).

• Ordered List

```
<ol>
  <li>First Item</li>
  <li>Second Item</li>
  <li>Third Item</li>
</ol>
```

Output:

1. First Item
2. Second Item
3. Third Item

• Unordered List

```
<ul>
  <li>Apple</li>
  <li>Banana</li>
  <li>Orange</li>
</ul>
```

Output:

- Apple
- Banana
- Orange

7. Links in HTML

Links allow users to navigate between pages.

```
<a href="page2.html">Go to Page 2</a>    (href specifies the URL)
```

8. Images in HTML

Images are added using the `<img>` tag.

```

```

src : Path of the image

alt : Alternate text (shown if image not found)

width : Width of the image (in pixels)

height : Height of the image (in pixels)

9. HTML Attributes

Attributes provide additional information about HTML elements.

```
<tag attribute="value">Content</tag>
```

Example: `<a href="https://example.com" target="_blank">Visit Site</a>`

`target="_blank"` opens the link in a new tab.

5. Attributes in HTML

Attributes provide additional information about HTML elements.

Attribute	Description	Example
id	Unique identifier for an element	<code><p id="intro"> Hello </p></code>
class	Specifies one or more class names for an element	<code><p class="title"> Hello </p></code>
style	Defines inline CSS styles	<code><p style="color:blue;"> Hello </p></code>
href	Specifies the URL of a link	<code> Link </code>
src	Specifies the source URL of an image	<code></code>
alt	Provides alternative text for an image	<code></code>
title	Provides extra information about an element	<code><p title="This is a tooltip"> Hover over me </p></code>
target	Specifies where to open the linked document	<code> Visit </code>

6. Headings in HTML

HTML headings are defined with `<h1>` to `<h6>` tags.

`<h1>` Heading 1 (Largest) `</h1>`
`<h2>` Heading 2 `</h2>`
`<h3>` Heading 3 `</h3>`
`<h4>` Heading 4 `</h4>`
`<h5>` Heading 5 `</h5>`
`<h6>` Heading 6 (Smallest) `</h6>`

Note :

Use headings in proper order for better structure and SEO.

7. Paragraphs and Line Break

- `<p>` tag is used to define a paragraph.
- `<br>` tag is used to break a line.

Example :

`<p>`This is a paragraph.`</p>`
This is first line.`<br>`
This is second line.

8. Text Formatting in HTML

Tag	Description	Example
<code></code>	Bold text	<code></code> Bold Text <code></code>
<code></code>	Important text (bold)	<code></code> Important Text <code></code>
<code><i></code>	Italic text	<code><i></code> Italic Text <code></i></code>
<code></code>	Emphasized text (italic)	<code></code> Emphasized Text <code></code>
<code><u></code>	Underlined text	<code><u></code> Underlined Text <code></u></code>
<code><mark></code>	Highlighted text	<code><mark></code> Highlighted Text <code></mark></code>
<code><small></code>	Smaller text	<code><small></code> Small Text <code></small></code>

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10. HTML Elements

An HTML element is everything from the start tag to the end tag.

`<p>` This is a paragraph. `</p>` `<!-- p is the element -->`
Start Tag Content End Tag

11. Block vs Inline Elements

Block Level Elements	Inline Elements
• Start on a new line. • Take full width available. • Examples: <code><div></code>, <code><h1></code> to <code><h6></code>, <code><p></code>, <code><form></code>, <code></code>, <code></code>	• Do not start on a new line. • Take only required width. • Examples: <code></code>, <code><a></code>, <code></code>, <code></code>, <code><i></code>, <code><input></code>

Example :

`<h1>` Heading `</h1>`
`<p>` This is a paragraph. `</p>`
`<div>` This is a div. `</div>`

→ (Each element starts on a new line)

`<b>` Bold `</b>` `<i>` Italic `</i>`
`<span>` Span `</span>` `<a>` Link `</a>`

↓ (Elements are on the same line)

12. HTML Headings

HTML headings are defined with the `<h1>` to `<h6>` tags.

`<h1>` This is Heading 1 `</h1>`
`<h2>` This is Heading 2 `</h2>`
`<h3>` This is Heading 3 `</h3>`
`<h4>` This is Heading 4 `</h4>`
`<h5>` This is Heading 5 `</h5>`
`<h6>` This is Heading 6 `</h6>`

Output in Browser :

This is Heading 1
This is Heading 2
This is Heading 3
This is Heading 4
This is Heading 5
This is Heading 6

`<h1>` is the most important and biggest.
`<h6>` is the least important and smallest.

13. HTML Paragraph

The `<p>` tag defines a paragraph in HTML.

Output :

`<p>` This is a paragraph. `</p>`
`<p>` HTML is easy to learn. `</p>`

This is a paragraph.
HTML is easy to learn.

14. HTML Line Break and Horizontal Rule

★ Line Break: `<br>`

The `<br>` tag inserts a single line break.

First Line `<br>`
Second Line `<br>`
Third Line

Output :

First Line
Second Line
Third Line

★ Horizontal Rule: `<hr>`

The `<hr>` tag defines a horizontal rule.

Content Above
`<hr>`
Content Below

Output :

Content Above

Content Below

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15. HTML Attributes

Attributes provide additional information about HTML elements.

```
<tag attribute = "value" >Content </tag>
```

Example :

```
<a href = "https://example.com" target = "_blank" >Visit Site </a>
```

href = link destination

target = where to open
("_blank" opens in new tab)

16. HTML Text Formatting

Tag	Description	Example	Output
	Bold text	 Bold Text 	Bold Text
<i>	Italic text	<i> Italic Text </i>	<i>Italic Text</i>
<u>	Underlined text	<u> Underlined Text </u>	<u>Underlined Text</u>
	Important text (bold)	 Important 	Important
	Emphasized text (italic)	 Emphasized 	<i>Emphasized</i>
<mark>	Marked / highlighted text	<mark> Highlight </mark>	Highlight
<small>	Smaller text	Small Text	Small Text
	Deleted text	 Deleted 	Deleted
<ins>	Inserted text	<ins> Inserted </ins>	<u>Inserted</u>

17. HTML Lists Summary

Ordered List ()	Unordered List ()	Description List (<dl>)
• Items are in numbered order. • Uses and tags. Example : <pre> HTML CSS JavaScript </pre>	• Items are in bullet points. • Uses and tags. Example : <pre> Apple Banana Orange </pre>	• Used for terms and descriptions. • Uses <dl>, <dt>, <dd> tags. Example : <pre><dl> <dt> HTML </dt> <dd> HyperText Markup Language </dd> <dt> CSS </dt> <dd> Cascading Style Sheets </dd> </dl></pre>

18. HTML Links

The <a> tag defines a hyperlink.

Example Code	Description	Output
 Google 	Link to another website	<u>Google</u>
 Go to Page 2 	Link to another page	<u>Go to Page 2</u>
 Go to Section 	Link to a section on same page	<u>Go to Section</u>
 Email Us 	Link to send an email	<u>Email Us</u>

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19. HTML Images

Images are added using the `<img>` tag.

```

```

- `src` : Specifies the path of the image.
- `alt` : Provides alternate text if image is not found.
- `width` : Sets the width of the image (in pixels).
- `height` : Sets the height of the image (in pixels).

Example Output :



20. HTML Tables

Tables are used to arrange data in rows and columns.

```
<table border="1">
<tr>
  <th> Roll No. </th> <th> Name </th> <th> Course </th>
</tr>
<tr>
  <td> 1 </td> <td> Sagar </td> <td> BCA </td>
</tr>
<tr>
  <td> 2 </td> <td> Rahul </td> <td> MCA </td>
</tr>
</table>
```

Output in Browser :

Roll No.	Name	Course
1	Sagar	BCA
2	Rahul	MCA

Common Tags Used in Table :

- `<table>` : Defines the table
- `<tr>` : Table row
- `<th>` : Table header (bold & centered)
- `<td>` : Table data (cell)
- `border` : Adds border to the table

21. HTML Forms

Forms are used to collect user input.

```
<form action="submit.php" method="post">
Name : <input type="text" name="name"><br><br>
Email : <input type="email" name="email"><br><br>
Course : <select name="course">
  <option value="bca">BCA</option>
  <option value="mca">MCA</option>
</select><br><br>
<input type="submit" value="Submit">
</form>
```

Output in Browser :

Name :

Email :

Course : ▼

Common Input Types :

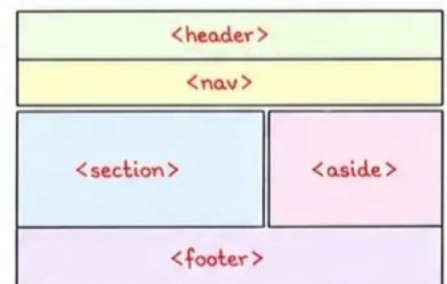
- `text` : Single line text
- `email` : Email field
- `number` : Number field
- `radio` : Radio button
- `checkbox` : Checkbox
- `submit` : Submit button
- `reset` : Reset button

22. HTML Semantic Elements

Semantic elements clearly describe their meaning to both browser and developer.

Example Structure :

Semantic Tag	Purpose
<code><header></code>	Defines header for a document or section
<code><nav></code>	Defines navigation links
<code><section></code>	Defines a section in a document
<code><article></code>	Defines independent, self-contained content
<code><footer></code>	Defines footer for a document or section
<code><aside></code>	Defines content aside from the content (like sidebar)



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23. HTML Div and Span

- `<div>` is a block level element.
- `<span>` is an inline element.

`<div>` Example (Block Level)

```
<div style="border:1px solid blue;
padding:10px; margin:5px 0;">
This is a div. It takes full width.
</div>
```

(Starts on new line and takes full width)

`<span>` Example (Inline)

This is `<span style="color:red;">`
a span`</span>` inside a line.

(Does not start on new line
and takes only required width)

24. HTML Comments

Comments are not displayed in the browser. They are used to add notes in code.

```
<!-- This is a single line comment -->
<!--
  This is
  a
  multi-line comment.
-->
```

★ Note :

Comments help in understanding
the code and are ignored
by the browser.

25. HTML Colors

Colors can be specified using color names, HEX values, RGB values.

Method	Example	Code	Result
1. Color Name	Red	<code><p style="color:red;">This is Red Text</p></code>	This is Red Text
2. HEX Value	#1E90FF	<code><p style="color:#1E90FF;">This is Blue Text</p></code>	This is Blue Text
3. RGB Value	rgb(0,128,0)	<code><p style="color:rgb(0,128,0);">This is Green Text</p></code>	This is Green Text

26. HTML List Types

Ordered List (ol)	Unordered List (ul)	Description List (dl)
• Items are numbered. • Uses <code></code> and <code></code> tags. Example : <pre> HTML CSS JavaScript </pre> Output : <pre>1. HTML 2. CSS 3. JavaScript</pre>	• Items are in bullet points. • Uses <code></code> and <code></code> tags. Example : <pre> Apple Banana Orange </pre> Output : <pre>• Apple • Banana • Orange</pre>	• Used for terms & descriptions. • Uses <code><dl></code>, <code><dt></code>, <code><dd></code> tags. Example : <pre><dl> <dt>HTML</dt> <dd>HyperText Markup Language</dd> <dt>CSS</dt> <dd>Cascading Style Sheets</dd> </dl></pre>

27. HTML Doctype Declaration

The `<!DOCTYPE>` declaration defines the document type and version of HTML.

`<!DOCTYPE html>` This declaration must be the first line of an HTML document.

28. Basic HTML Document Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>My First Web Page</title>
</head>
<body>
  <h1>Welcome to HTML!</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

→ `<!DOCTYPE html>` Defines HTML5 document
→ `<html>` Root element of HTML page
→ `<head>` Contains meta information
→ `<title>` Title of the page (shown in browser tab)
→ `<body>` Contains visible content of the page
→ `<h1>` Heading (most important)
→ `<p>` Paragraph

Browser Output :

